

MUHAMMED MUIZ ARUMMAL

AI Engineer

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PROFILE

AI engineer who builds and evaluates LLM and multi-agent systems — from the data pipeline through to a deployed product. MSc AI & Machine Learning at the University of Birmingham, based in the UK, with a security background (CompTIA CySA+). I focus on whether an AI system actually works in the real world, not just in a demo: building the product and the tests that prove it.

EDUCATION

University of Birmingham — MSc, Artificial Intelligence & Machine Learning Sept 2026 – Sept 2027 (in progress)

Modules: Neural Computation, Natural Language Processing, Evolutionary Computation.

TKM Institute of Technology — B.Tech, Computer Science & Engineering 2021 – 2025

RESEARCH EXPERIENCE

Testing AI attacker agents against a multi-agent app github.com/seclayer/finbot-bench

- Built a benchmark where local LLMs act as autonomous attackers against a deliberately vulnerable multi-agent AI application (**OWASP FinBot CTF**), scored by the app's own exploit detector rather than the attacker marking its own work.
- Compared three attacker designs — one-shot, a feedback loop, and a team of cooperating agents — and found the attacker's *design* mattered more than the size of the model behind it.

Are AI safety scores measuring the right thing? preprint in preparation | github.com/seclayer/jlens-fuzz

- Showed that a widely reused automated judge, which decides whether an attack on a chatbot “worked”, rewards replies that merely *look* like a jailbreak over ones that are actually harmful — so published success rates built on it are inflated.
- Built a stricter replacement judge and reported honestly where it still fails, rather than claiming a clean fix.

Independent reproduction of a 2026 jailbreak paper github.com/seclayer/reproductions

- Rebuilt the paper's method end to end on three local models, tested a defence the original authors named but never evaluated, and also reproduced **GPTFuzzer**.
- Caught the automated grader scoring wrongly in both directions, re-checked every result by hand, and published the corrections next to the originals.

Security audit of a deployed multi-agent platform authorization review & fix

- Reviewed a live agent-to-agent platform against a published threat checklist, found a missing access-control check, fixed it, and proved the fix with an automated test.

PROJECTS

Multi-agent AI assistant platform (GptGram) github.com/seclayer/Gptgram-production-ready

- Built and deployed an AI platform on **AWS** where several **AI agents** work together to answer a question, with a **retrieval (RAG)** layer on **Qdrant** so answers are grounded in real documents, and conversation state saved to **S3**.
- Ran it under real load with background job queues (**Celery**) and **REST APIs**, so many people can use it at once without slowing down.

AI system for security monitoring (Secure Guard) github.com/seclayer/ai-soc

- Built a data pipeline (Sysmon → Logstash → **Elasticsearch**, cleaned with **Python** and **SQL**) that turns messy raw security logs into usable data.
- Trained **XGBoost** and **SVM** models to flag likely malware, and used **Prophet** forecasting to turn alert trends into clear response steps, so analysts see real threats first.

Custom GPU kernels (personal project) CUDA | cuTile

- Wrote my own **CUDA** kernels to reverse-engineer PyTorch's operations and try to reach its speed, and returned to it with **cuTile** when it was released. Did not surpass PyTorch.

Also built: **Ggram**, an agentic platform that uses **graph neural networks** to route and solve ML tasks; a **scikit-learn** security chatbot that cut false alarms in a simulated SOC by combining several models (proof of concept, not public).

TECHNICAL SKILLS

- **Languages:** Python, C++ (incl. multithreading), SQL; CUDA (incl. cuTile) GPU programming.
- **AI / ML:** pandas, NumPy, scikit-learn, PyTorch, TensorFlow, JAX, XGBoost; generative AI & LLMs, agentic & multi-agent systems, RAG, prompt engineering, model evaluation, LLM red-teaming, prompt injection.
- **Cloud & data:** AWS (EC2, S3), Docker, Linux, Git / CI-CD, REST APIs, network programming, Databricks, Power BI; Agile / Scrum.
- **Tools / frameworks:** LangChain, Qdrant, Ollama, Celery; Grafana, Prometheus.
- **Professional:** clear technical writing for non-specialists, honest reporting of results, teamwork on undergraduate team projects.
- **Security:** detection engineering, SIEM (ELK, Splunk), incident response, threat hunting, MITRE ATT&CK.

CERTIFICATIONS

- **CompTIA CySA+ (CS0-003)** — [verify](#) | **Mathematics for Machine Learning**, Imperial College London (Coursera) — [verify](#) | **SOC Analyst**, TryHackMe — [certificate](#)
- **In progress:** AWS Certified Generative AI Developer (AIP-C01) — prep course completed, exam not yet sat | HTB Certified Offensive AI Expert (COAE) — in progress.

Also: Kaggle competitor (ARC Prize 2026, ARC-AGI-3); ACM chapter participation.